Practical 7

```
sudo docker start oracle-xe
Sudo docker ps –a
Docker start container id/
sudo docker exec -it oracle-xe sqlplus system/system123
```

```
CREATE TABLE Library (
    book_id NUMBER,
    book_name VARCHAR2(50),
    author VARCHAR2(50),
    category VARCHAR2(50),
    date1 DATE,
    operation VARCHAR2(50)
);

CREATE TABLE Library_Audit (
    book_id NUMBER,
    book_name VARCHAR2(50),
    author VARCHAR2(50),
    category VARCHAR2(50),
    date1 DATE,
    operation VARCHAR2(50)
);

INSERT INTO Library VALUES (101, 'Book1', 'Author1', 'Fiction', TO_DATE('01-
01-2026','DD-MM-YYYY'), 'Issue');
INSERT INTO Library VALUES (102, 'Book2', 'Author2', 'Fiction', TO_DATE('02-
01-2026','DD-MM-YYYY'), 'Return');
INSERT INTO Library VALUES (103, 'Book3', 'Author3', 'History', TO_DATE('03-
01-2026','DD-MM-YYYY'), 'Issue');
INSERT INTO Library VALUES (104, 'Book4', 'Author4', 'Fiction', TO_DATE('04-
01-2024','DD-MM-YYYY'), 'Return');
INSERT INTO Library VALUES (105, 'Book5', 'Author5', 'Science', TO_DATE('05-
01-2024','DD-MM-YYYY'), 'Issue');
INSERT INTO Library VALUES (106, 'Book6', 'Author6', 'History', TO_DATE('06-
01-2024','DD-MM-YYYY'), 'Return');
INSERT INTO Library VALUES (107, 'Book7', 'Author7', 'Fiction', TO_DATE('07-
01-2024','DD-MM-YYYY'), 'Issue');
INSERT INTO Library VALUES (108, 'Book8', 'Author8', 'Science', TO_DATE('08-
01-2024','DD-MM-YYYY'), 'Return');
INSERT INTO Library VALUES (109, 'Book9', 'Author9', 'History', TO_DATE('09-
01-2024','DD-MM-YYYY'), 'Issue');
INSERT INTO Library VALUES (110, 'Book10', 'Author10', 'Fiction',
TO_DATE('10-01-2024','DD-MM-YYYY'), 'Return');
```

```
CREATE OR REPLACE TRIGGER upd_library
AFTER UPDATE ON Library
FOR EACH ROW
BEGIN
    INSERT INTO Library_Audit
```

```
    VALUES (
        :OLD.book_id,
        :OLD.book_name,
        :OLD.author,
        :OLD.category,
        SYSDATE,
        'Book updating'
    );
END;
/


UPDATE Library SET book_id = 111 WHERE book_name = 'Book1';



CREATE OR REPLACE TRIGGER del_library
AFTER DELETE ON Library
FOR EACH ROW
BEGIN
    INSERT INTO Library_Audit
    VALUES (
        :OLD.book_id,
        :OLD.book_name,
        :OLD.author,
        :OLD.category,
        SYSDATE,
        'Deleting'
    );
END;
/


DELETE FROM Library WHERE book_name = 'Book2';

SELECT * FROM Library_Audit;
```